

[JPJC/2025/Q2]

- 2 When a user visits webpages, their browsing history is stored as a linked list in chronological order (oldest to newest). Additionally, the system supports Back and Forward navigation using two stacks, both implemented using arrays.

- The Back Stack stores the pages the user navigated. The top of Back Stack is the current page navigated.
- The Forward Stack stores the pages popped when the user presses "Back", and these can be restored using the "Forward" button.

The class `PageNode` contains two properties:

- `url` is the string data in the node
- `next` points to the next node.

The class `BrowserHistory` manages a user's browsing history as a linked list. When the user visits a webpage, a `PageNode` object is inserted. It contains one property:

- `head` points to the first node.

The class `BrowserHistory` contains the following methods:

- a constructor to set the head pointer to `None`
- `visit_page(url)` to add the node with `url` data to the end of the linked list
- `print_history()` to output all `url` data from first visited to last visited.

Task 2.1

Write program code to declare the class `PageNode` and its constructor.

Write the program code to declare the class `BrowserHistory` and its methods.

[8]

The class `BrowserNavigator` contains the following properties:

- `current_page` is the URL string of the currently viewed page
- `back_stack` is the stack array that stores up to 10 URL strings of pages visited before the current one
- `forward_stack` is the stack array that stores up to 10 URL strings of pages that were navigated away from by going back
- `back_top` is the top pointer of the back stack
- `forward_top` is the top pointer of the forward stack.

The class `BrowserNavigator` contains the following methods:

- a constructor to set the properties to appropriate `NULL` values
- `go_back()` to push the current page onto the forward stack, then pop the top page from the back stack and assign it to the current page
- `go_forward()` to push the current page onto the back stack, then pop the top page from the forward stack and assign it to the current page
- `visit_new_page(url)` to push the current page onto the back stack, clear the forward stack, and then assign the URL string to the current page.

[JPJC/2025/Q2 Cont']

Task 2.2

Write the program code to declare the class `BrowserNavigator` and its methods. [10]

Task 2.3

Write the main program to:

- declare a new instance of `BrowserHistory`
- declare a new instance of `BrowserNavigator`
- update the browser history and browser navigator when the user
 - visits `"jpjc.coursemology.org"`, `"jpjc.moe.edu.sg"`, and followed by `"web.whatsapp.com"`
 - goes back twice
 - goes forward once
- output the URL of the current page
- output the URLs of the full browsing history. [6]

Save your Jupyter Notebook for Task 2.